



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)

REVIEW TEST

21-02-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

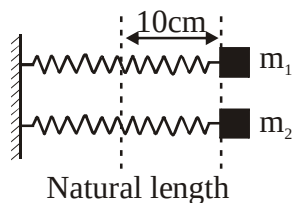
SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A rock with mass $m = 3 \text{ kg}$ is suspended from the roof of an elevator by a light cord. The rock is totally immersed in a bucket of water that is kept on the floor of the elevator but the rock doesn't touch the bottom or sides of the bucket. When the elevator is at rest tension in the cord is 21N. Now the lift is accelerated with an acceleration $3\sqrt{2} g$ at an angle 45° above the horizontal, tension in the string becomes : (Rock still doesn't touch the bucket).
- (A) 84 N (B) 105 N (C) $63\sqrt{2} \text{ N}$ (D) 63 N

Space for Rough Work

2. A plastic circular disc of radius R is placed on a thin oil film, spread over a flat horizontal surface. The torque required to spin the disc about its central vertical axis with a constant angular velocity is proportional to
- (A) R^2 (B) R^3 (C) R^4 (D) R^6
3. The drawing shows a top view of a frictionless horizontal surface, where there are two identical springs with particles of mass m_1 and m_2 attached to them. Each spring has a spring constant of 1800 N/m . The particles are pulled to the right and then released from the positions shown in the drawing. How much time passes before the particles are again side by side for the first time if $m_1 = 2 \text{ kg}$ and $m_2 = 8 \text{ kg}$?



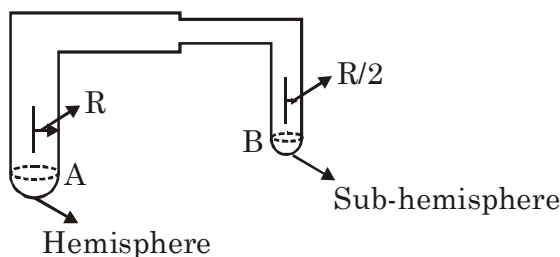
- (A) $\frac{\pi}{45} \text{ sec}$ (B) $\frac{2\pi}{45} \text{ sec}$ (C) $\frac{3\pi}{45} \text{ sec}$ (D) $\frac{4\pi}{45} \text{ sec}$

Space for Rough Work

4. Time period of a physical pendulum about one pivot point A is T. There is another pivot point B on opposite side of the centre of mass which gives the same time period. Pivot A, centre of mass and Pivot B are in same line. Distance between A and B is L. Choose the correct relation.

(A) $g = \frac{L}{T^2}$ (B) $g = \frac{L}{2\pi T^2}$ (C) $g = \frac{2\pi L}{T^2}$ (D) $g = L\left(\frac{2\pi}{T}\right)^2$

5. Two tubes A and B of radii R and $\frac{R}{2}$ respectively are connected to each other as shown in figure. At one end of tube A, there is a hemispherical bubble of soap solution of surface tension S and on the end of tube B there is a sub-hemispherical bubble of soap solution of surface tension S'. From this one can conclude that



(A) $S' = \frac{S}{2}$ (B) $S' < \frac{S}{4}$ (C) $S' > \frac{S}{2}$ (D) $\frac{S}{4} < S' < \frac{S}{2}$

Space for Rough Work

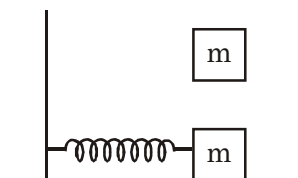
SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
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Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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6. A 20 gm particle is subjected to two simple harmonic motions $x_1 = 2 \sin 10t$, $x_2 = 4 \sin \left(10t + \frac{\pi}{3} \right)$. Where x_1 and x_2 are in metre and t is in sec.
- (A) The displacement of the particle at $t = 0$ will be $2\sqrt{3}$ m
- (B) Maximum speed of the particle will be $20\sqrt{7}$ m/s
- (C) Magnitude of maximum acceleration of the particle will be $200\sqrt{7}$ m/s².
- (D) Energy of the resultant motion will be 28 J.
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Space for Rough Work

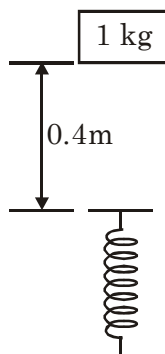
7. The potential energy of a particle of mass 0.1 kg moving along the x-axis is given by $U = 5x(x-4)$ J, where x is in meters. It can be concluded that:
 (A) the particle is acted upon by a constant force
 (B) the speed of the particle is maximum at $x = 2$ m
 (C) the particle executes simple harmonic motion
 (D) the period of oscillation of the particle is $\pi/5$ seconds
8. A particle of mass M is performing SHM with amplitude A. When the particle was at $A/2$ and moving away from mean position, a block of same mass was kept on it such that it sticks to it.



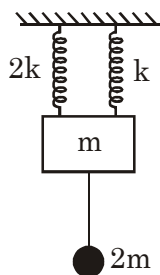
- (A) The new amplitude $= \sqrt{\frac{5}{8}}A$ (B) The time period remains unchanged
- (C) The new amplitude $= \sqrt{\frac{7}{16}}A$ (D) The time period increases
9. Three simple harmonic motions in the same direction each having amplitude "a" and the same time period, are superposed. If each differs in phase from the next by $\frac{\pi}{3}$ then
 (A) Resultant amplitude is $2a$
 (B) Phase of resultant motion relative to first is $\frac{\pi}{3}$
 (C) The energy associated with the resulting motion is 4 times the energy associated with any single motion
 (D) Maximum speed of resultant SHM will be more than double of the initial SHM's

Space for Rough Work

10. A block of mass $m = 1 \text{ kg}$ falls from a height $h = 0.4 \text{ m}$ on a massless spring of stiffness constant $k = 300 \text{ Nm}^{-1}$. If $g = 10 \text{ m/s}^2$, then



- (A) when compression in the spring is 0.1 m , the acceleration of the block is zero
 (B) maximum compression in the spring is 0.2 m
 (C) maximum acceleration of the block is 50 m/s^2
 (D) maximum extension in the spring is 0.2 m
11. A bob of mass $2m$ hangs by a string attached to the block of mass m of a spring block system. The whole arrangement is in a state of equilibrium. The bob of mass $2m$ is pulled down slowly by a distance x_0 and released.



- (A) For $x_0 > \frac{mg}{k}$, minimum tension in string is mg
 (B) For $x_0 = \frac{mg}{k}$ maximum tension in string is $4mg$
 (C) The motion will remain simple harmonic for $x_0 \leq \frac{mg}{k}$
 (D) Frequency of oscillation of system is $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$, for all non-zero values of x_0

Space for Rough Work

12. The diagram below shows a cross-sectional view of a cylindrical pipe of varying diameter. If an ideal fluid is flowing through the pipe, find the true statement(s).



- (A) Volume flow rate is same at A & B
 (B) The pressure is lower at point B than at point A.
 (C) The volume flow rate is greater at point A than at point B.
 (D) The flow speed is greater at point B than at point A.
13. Water jet coming out of a stationary horizontal tube at speed v strikes horizontally a massive wall moving in opposite direction with same speed. Water comes to rest relative to wall after striking. Treating A as cross-section of jet and density of water as ρ . Select the correct alternative(s)
- (A) Force exerted on the wall is $2\rho Av^2$
 (B) Force exerted on the wall is $4\rho Av^2$
 (C) Rate of change of kinetic energy of water jet striking the wall is $8\rho Av^3$
 (D) Rate of change of kinetic energy of water jet striking the wall is zero.

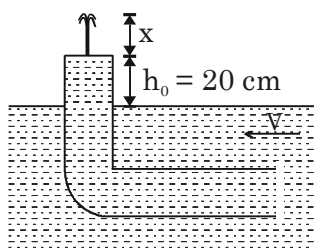
Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)
 No question will be asked in section II

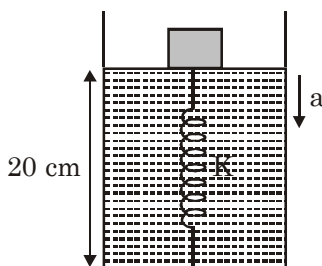
SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

1. A bent tube is lowered into water stream as shown in the figure. The velocity of the stream relative to the tube is equal to $V = 4 \text{ m/s}$. The closed upper end of the tube located at height $h_0 = 20 \text{ cm}$ above free surface of water, has a small orifice. The height h (in cm) to which the water will get spurt is x . Value of $x/10$ is

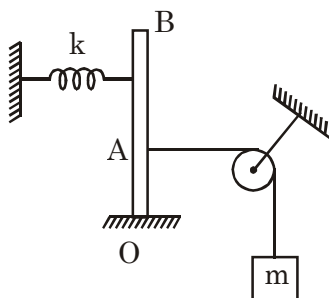


2. A rectangular tank having base $15\text{cm} \times 20\text{cm}$ is filled with water (density $\rho = 1000 \text{ kg m}^{-3}$) upto 20 cm height. One end of an ideal spring of natural length $h_0 = 20 \text{ cm}$ and force constant $K = 280 \text{ Nm}^{-1}$ is fixed to the bottom of a tank so that spring, remains vertical. This system is in an elevator moving downwards with acceleration $a = 2 \text{ ms}^{-2}$. A cubical block of side $\ell = 10 \text{ cm}$ and mass $m = 2 \text{ kg}$ is gently placed over the spring and released gradually, as shown in figure. If block is slightly pushed down from equilibrium position and released, and the frequency of its vertical oscillations is $\frac{a\sqrt{2}}{\pi}$, then calculate a .

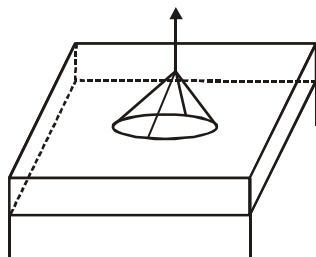


Space for Rough Work

- 3 A massless rod is hinged at O. String carrying a mass m at one end is attached to point A on the rod so that $OA = a$. At another point B ($OB = b$) of the rod a horizontal spring of force constant k is attached as shown. Find the period of small vertical oscillations of mass m around its equilibrium position. (If $m = 10\text{ kg}$, $k = 10\text{ N/m}$, $a = 1\text{ m}$, $b = 2\pi\text{ m}$)



4. Particles of sand are sprinkled on to the surface of water in a beaker, filled to a depth of 45.45 mm to reach the bottom. The sand particles are spherical and have a diameter of 0.10 mm . The beaker is kept in a lift moving with an upward acceleration of 1.2 m/s^2 . Estimate the minimum time to reach the bottom by the particles if viscosity of water $\eta = 1.1 \times 10^{-3}\text{ Nsm}^{-2}$, density of water is 1000 kg/m^3 , density of sand is 2000 kg/m^3 and g is 9.8 ms^{-2} . (Express your answer in second).
5. The surface tension of water is 75 dyne/cm . A ring is initially resting on a horizontal water surface as shown in the figure. The circumference of the ring is 20 cm and its weight is 0.1 N . The minimum vertical force (in N) required to pull the thin wire ring up is x . Value of $100x/13$ is



Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 15)

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1. Identify the chemical specie in which central atom is sp^2 hybridised and unpaired electron reside in non-hybridised orbital.

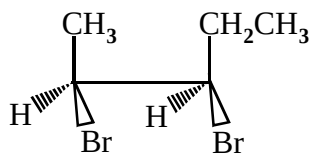
- (A) ClO_2 (B) $\dot{C}F_3$ (C) $\dot{C}HF_2$ (D) NO_2



- (A) I and II are identical (B) I and II are enantiomer
 (C) I and II are diastereomer (D) I and II are structural isomer
3. Which of the following crystalline systems exists as only two Bravais lattices ?
 (A) Monoclinic (B) Hexagonal (C) Orthorhombic (D) Rhombohedral

Space for Rough Work

4. Which of the following characteristics does not show similarity between benzene and borazine?
 (A) Total number of electron (B) Aromatic character
 (C) Presence of $2p\pi$ - $2p\pi$ bond (D) Polarity of bond present in ring
5. Correct name of following compound is



- (A) 2(R), 3(R)–Dibromopentane (B) 2(S), 3(S)–Dibromopentane
 (C) 2(R), 3(S)–Dibromopentane (D) 2(S), 3(R)–Dibromopentane

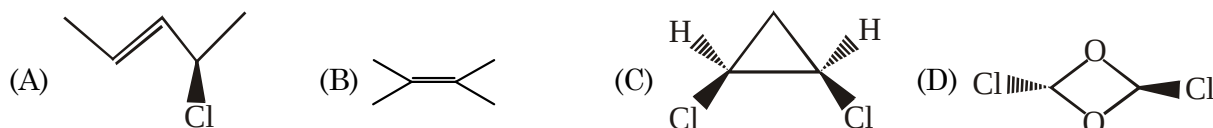
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

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6. In rock salt structure which of the following is always true
 [Given : a = edge length of unit cell r_+ = radius of cation, r_- = radius of anion]
- (A) One cation is surrounded by six nearest anions & vice-versa
- (B) $\frac{a}{2} = r_+ + r_-$
- (C) $\sqrt{2}a = 4r_-$
- (D) $\frac{r_+}{r_-} = \sqrt{2} - 1$
7. Which of the following order is/are not correct?
- (A) $H^- > F^-$ (Radius) (B) $Na < Be$ (2^{nd} IE)
- (C) $N > P$ (Electron affinity) (D) $F < Cl$ (Electronegativity)

Space for Rough Work

8. Identify compound which can show geometrical isomer as well as optical isomerism :



9. Select the correct statement(s)-

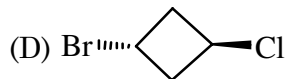
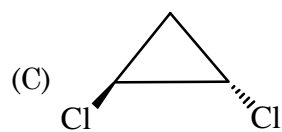
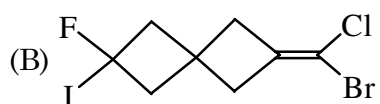
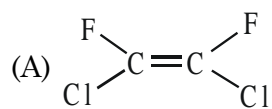
- (A) In cubic close packing each atom is surrounded by six octahedral voids.
 (B) In cubic close packing each atom is surrounded by eight tetrahedral voids.
 (C) In simple cubic crystal, each atom is surrounded by eight cubical voids
 (D) In hexagonal close packing, each atom is surrounded by four tetrahedral voids

10. Which of the following reaction is/are **CORRECT**?

- (A) $B_2H_6 + 2NaH \longrightarrow 2Na[BH_4]$
 (B) $B_2H_6 + 2NH_3 \longrightarrow [BH_2(NH_3)_2]^+ [BH_4]^-$
 (C) $B_2H_6 + 2CO \longrightarrow 2(BH_3 \cdot CO)$
 (D) $B_2H_6 + MeNH_2 \longrightarrow 2(BH_3 \cdot MeNH_2)$

Space for Rough Work

11. Which of the following is/are not optically active?



Space for Rough Work

12. A metal exists in two allotropic forms, α -form is BCC and β -form is FCC. The distance between nearest neighbours is same in both forms. Which of the following is/are **INCORRECT** information(s) related with these two forms of the metal ?
- (A) Next nearest neighbours are at same distance in both forms.
(B) Percentage void space in α -form is greater than that in β -form.
(C) The diameter of largest sphere which may be fitted in both the crystals of metal, without disturbing the distance between nearest neighbours, is same in both forms.
(D) The edge length is greater for the unit cell of β -form.
13. The chemical specie(s) which does not exist.
- (A) PH_5 (B) PbI_4 (C) BiI_5 (D) PbCl_4

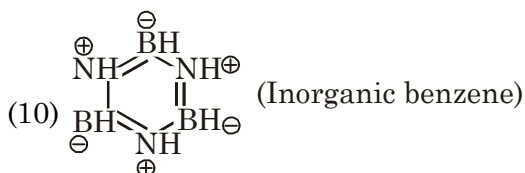
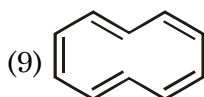
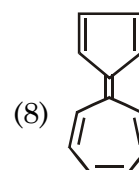
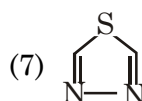
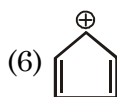
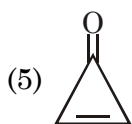
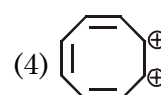
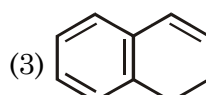
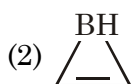
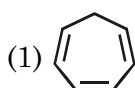
Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)
No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
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1. How many of following compound are aromatic in nature ?



2. In diamond carbon atoms occupy FCC lattice points as well as alternate tetrahedral voids (structure similar to zinc blende). Find the packing fraction of the diamond unit cell

$$\left[\text{Given : } \pi = \frac{5.44}{\sqrt{3}} \right]$$

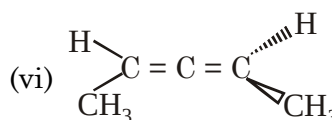
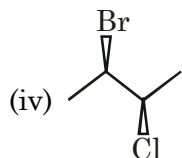
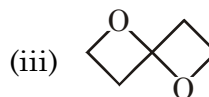
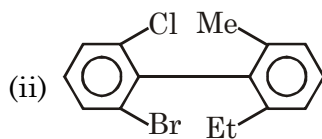
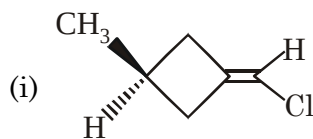
First multiply your answer with 100. Then fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

Space for Rough Work

3. Find the number of non planar chemical species in which multicentered bond (3c-2e or 3c-4e) is present.



4. How many chiral molecules are present among the following compounds?



5. If zinc blende (ZnS) crystal in which sulphide ions occupy corners of the cubic unit cells is heated, all ions present on any two axis passing through opposite lying face centres are lost. Bivalent anion (P^{2-}) enters each unit cell to maintain electrical neutrality without disturbing other ions in lattice. If new formula of unit cell becomes $\text{Zn}_x\text{S}_y\text{P}_z$, then $(x + y + z)$ is.

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 15)**

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Number of solutions of equation $2020\text{sgn}(x^3 - 4x) + 2022\text{sgn}(x^2 - x) = 2021$, is
(sgn denotes signum function)

(A) 0 (B) 2 (C) 5 (D) infinite

2. Range of the function $f(x) = \frac{3\sin x + 4\cos x - 2}{3\sin x + 4\cos x + 10}$, is

(A) $\left[0, \frac{1}{5}\right]$ (B) $\left[-1, \frac{1}{5}\right]$ (C) $[-2, 1]$ (D) $\left[-\frac{7}{5}, \frac{1}{5}\right]$

Space for Rough Work

3. Sum of all integral values of x which are in the domain of function $f(x) = \sqrt{\sin x} + \sqrt{x+1} + \sqrt{6-x}$, is
 (A) 5 (B) 6 (C) 7 (D) 20
4. Which of the following pair of functions are identical functions- (sgn denotes signum function)
 (A) $f(x) = \sqrt{1 + \sin 2x}$, $g(x) = \sin x + \cos x$ (B) $f(x) = \frac{\tan 2x - \tan x}{1 + \tan 2x \tan x}$, $g(x) = \tan x$
 (C) $f(x) = \operatorname{sgn}(x)$, $g(x) = \operatorname{sgn}(e^x - e^{-x})$ (D) $f(x) = \frac{\sin x}{\cos x}$, $g(x) = \frac{\sec x}{\operatorname{cosec} x}$
5. If function $f(x)$ satisfies relation $f(x-y) + f(x) + 8x + 2xy = f(y) + 2x^2 + 8y \forall x, y \in \mathbb{R}$, then $f(5)$ is equal to -
 (A) 0 (B) 10 (C) 15 (D) 5

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
6. In $\triangle ABC$, if $B(0, 0)$, $AB = BC$ and equation of side BC is $y = x$. If mid point of median BE (E lies on side AC) is $(1, 2)$, then
- (A) equation of side AB is $y = 7x$
(B) equation of side AB is $7y = x$
(C) centroid of $\triangle ABC$ has co-ordinates $\left(\frac{4}{3}, \frac{8}{3}\right)$
(D) area of $\triangle ABC$ is $\frac{20}{3}$ sq. units
-

Space for Rough Work

7. If a function $f(x)$ satisfies condition $f(4+x) + f(-x) = 0 \forall x \in \mathbb{R}$ & equation $f(x) = 0$ has exactly 7 roots, then
 (A) graph of $f(x)$ is symmetric about line $x = 1$
 (B) graph of $f(x)$ is symmetric about point $(2, 0)$
 (C) sum of all 7 roots of $f(x) = 0$ is 7
 (D) sum of all 7 roots of $f(x) = 0$ is 14
8. Two opposite sides of rhombus are $y = x - 2$ & $y = x + 4$. If one vertex of rhombus is $P(1, -1)$ & angle at this vertex is 45° , then co-ordinates of vertex opposite to P can be
 (A) $(-1, 3)$ (B) $(1 + 3\sqrt{2}, 5 + 3\sqrt{2})$
 (C) $(5, 9)$ (D) $(1 - 3\sqrt{2}, 5 - 3\sqrt{2})$
9. Consider pair of lines $P_1 : mx^2 - 2nxy - my^2 = 0$ & $P_2 : x^2 - 4nxy - y^2 = 0$. If P_2 is angle bisectors of P_1 & intercept made by P_1 on the line $x = 1$ is $2\sqrt{5}$, then $(m, n \neq 0)$
 (A) $m = -\frac{1}{8}$ (B) $m = \frac{1}{8}$ (C) $n = \frac{1}{4}$ (D) $n = -\frac{1}{4}$

Space for Rough Work

10. If lines $ax + by + c = 0$, $bx + cy + a = 0$ & $cx + ay + b = 0$ are concurrent (a, b, c are distinct non zero numbers), then $2^{a^2b^{-1}c^{-1}} \cdot 2^{b^2c^{-1}a^{-1}} \cdot 2^{c^2b^{-1}a^{-1}}$ is equal to
- (A) 8^{a+b+c} (B) $8^{\frac{a+b}{-c}}$ (C) $\left(\frac{c^2 - a^2 - b^2}{ab}\right)^3$ (D) $\frac{a^2 + b^2 + c^2}{ab + bc + ca}$
11. Let $f : [0, 4] \rightarrow [1, 33]$ is a polynomial of least degree such that $f(0) = 1$, $f(1) = 3$, $f(3) = 7$ & $f(4) = 33$, then
- (A) $f(x)$ is a one-one function (B) $f(x)$ is a many-one function
(C) $f(x)$ is a bijective function (D) $f(x)$ is a surjective function
12. $f : [4, \infty) \rightarrow [1, \infty)$ be an invertible function defined implicitly by relation $4^y + 4 = x2^y$, then
- (A) $f(x) = \log_2 \left(\frac{x - \sqrt{x^2 - 16}}{2} \right)$ (B) $f^{-1}(x) = 2^x + 4(2^{-x})$
(C) $f^{-1}(x) = 4(2^x) + 2^{-x}$ (D) $5f\left(\frac{5}{2}f(5)\right) = 2f^{-1}\left(\frac{2}{5}f^{-1}(2)\right)$
13. Let $f(x) = \max\{6|x| - x^2, |x|\}$, then equation $f(x) = k$ has
- (A) unique solution, if $k = 0$ (B) two solutions, if $k \in (0, 5)$
(C) four solutions, if $k = 5$ or $k = 9$ (D) six solutions, if $k \in (5, 9)$

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)
No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

-
1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x^3 + x^2 + x - 36$, then solution of equation $f(x) = f^{-1}(x)$, is
 2. Number of values of x for which $f \circ g(x) = g \circ f(x)$ where $f(x) = \frac{10^x - 1}{10^x + 1}$ & $g(x) = x + 1$, is
-

Space for Rough Work

3. If line $ax + by - 1 = 0$ intercepted between lines $ax + y + 1 = 0$ & $x + by = 0$ subtends 90° at origin ($a, b \in \mathbb{R}$), then maximum possible value of $24a$, is
4. Let $f : [1, \infty) \rightarrow \{2\} \rightarrow B$, $f(x) = 4 \frac{\sqrt{x-2}\sqrt{x-1}}{(2-x)}$ is an onto function then sum of all integers in set B , is
5. If $[x] + [2x] + [3x] + \dots + [2020x] = 1009\ell$, where $x = \frac{1010}{2021}$, then sum of digits of ℓ , is
(Where $[.]$ denotes greatest integer function)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	5	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	15
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
III	Single digit Integer (0-9)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator